

Q4-A: Provide examples of surfaces in $\mathbb{R}^3$ and explain their classification into types such as planes, spheres, and cylinders.

1 Introduction

A **surface** in $\mathbb{R}^3$ is a two-dimensional set of points satisfying an equation of the form

$$F(x, y, z) = 0,$$

where $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a given function. Surfaces are classified primarily by the degree of F and by their geometric properties (flatness, curvature, symmetry). The most important family is that of *quadric surfaces*, defined by second-degree polynomials in x, y, z .

2 Linear Surfaces: Planes

Definition 1 (Plane). A **plane** in $\mathbb{R}^3$ is the set of points (x, y, z) satisfying

$$ax + by + cz = d,$$

where $a, b, c, d \in \mathbb{R}$ and $(a, b, c) \neq (0, 0, 0)$. The vector $\mathbf{n} = (a, b, c)$ is called the *normal vector* to the plane.

Planes are the simplest surfaces: they have zero curvature everywhere and divide $\mathbb{R}^3$ into two half-spaces.

Example 2 (Coordinate planes).

- xy -plane: $z = 0$.
- yz -plane: $x = 0$.
- xz -plane: $y = 0$.

Example 3 (General plane). The plane passing through the origin with normal $\mathbf{n} = (1, 2, 3)$ is

$$x + 2y + 3z = 0.$$

3 Quadric Surfaces

Definition 4 (Quadric surface). A **quadric surface** is the solution set of a general second-degree equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0,$$

where the coefficients $A, B, \dots, J \in \mathbb{R}$ are not all zero. By a suitable rotation and translation of axes, every non-degenerate quadric can be reduced to one of the standard forms listed below.

3.1 Sphere

Definition 5 (Sphere). A **sphere** with centre $C = (x_0, y_0, z_0)$ and radius $r > 0$ is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2.$$

It is a quadric with equal positive coefficients on all three squared terms.

Key properties:

- Every cross-section by a plane is a circle (or a point, or empty).
- The sphere has constant positive Gaussian curvature $K = 1/r^2$.

Example 6. The sphere centred at the origin with radius 2:

$$x^2 + y^2 + z^2 = 4.$$

3.2 Ellipsoid

Definition 7 (Ellipsoid). An **ellipsoid** with semi-axes $a, b, c > 0$ has the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

When $a = b = c = r$, this reduces to a sphere of radius r .

Example 8.

$$\frac{x^2}{9} + \frac{y^2}{4} + \frac{z^2}{1} = 1$$

is an ellipsoid with semi-axes $a = 3, b = 2, c = 1$.

3.3 Cylinder

Definition 9 (Cylinder). A **(right circular) cylinder** aligned along the z -axis with radius $r > 0$ is

$$x^2 + y^2 = r^2.$$

More generally, any equation that is *independent of one variable* defines a cylindrical surface: the curve in the remaining two variables is *extruded* infinitely along the missing axis.

Key property: One eigenvalue of the associated matrix is zero, reflecting the lack of curvature along the axis direction.

Example 10 (Circular cylinder). $x^2 + y^2 = 9$ — radius-3 cylinder along the z -axis.

Example 11 (Parabolic cylinder). $y = x^2$ — a parabolic cylinder extending along the z -axis.

Example 12 (Elliptic cylinder). $\frac{x^2}{4} + \frac{y^2}{9} = 1$ — elliptic cross-section, extends along the z -axis.

3.4 Cone

Definition 13 (Elliptic cone). An **elliptic cone** with vertex at the origin is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

For $a = b$ this becomes a *right circular cone*.

Example 14. $x^2 + y^2 - z^2 = 0$, or equivalently $z = \pm\sqrt{x^2 + y^2}$, is a right circular cone with half-angle 45° .

3.5 Paraboloids

Definition 15 (Elliptic paraboloid). An **elliptic paraboloid** has the equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}.$$

Definition 16 (Hyperbolic paraboloid (saddle surface)). A **hyperbolic paraboloid** (saddle) has the equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Example 17. $z = x^2 + y^2$ is an elliptic paraboloid (a bowl opening upward).

Example 18. $z = x^2 - y^2$ is a hyperbolic paraboloid (a saddle at the origin).

3.6 Hyperboloids

Definition 19 (Hyperboloid of one sheet).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

This is a *connected* surface (one piece).

Definition 20 (Hyperboloid of two sheets).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1, \quad \text{equivalently} \quad \frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

This is a *disconnected* surface (two separate sheets).

Example 21. $x^2 + y^2 - z^2 = 1$: hyperboloid of one sheet (hourglass-like, connected).

Example 22. $z^2 - x^2 - y^2 = 1$: hyperboloid of two sheets (two separate bowls).

4 Classification Summary

Surface	Standard Equation	Key Feature
Plane	$ax + by + cz = d$	Degree 1; flat
Sphere	$x^2 + y^2 + z^2 = r^2$	Equal coefficients, $K > 0$
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	Generalised sphere
Circular Cylinder	$x^2 + y^2 = r^2$	Missing variable (z)
Elliptic Cylinder	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	Missing variable, elliptic cross-section
Parabolic Cylinder	$y = x^2$	Missing variable, parabolic cross-section
Cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$	Vertex at origin
Elliptic Paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	Opens in one direction
Hyperbolic Paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$	Saddle-shaped
Hyperboloid sheet)	(1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	Connected; waist circle
Hyperboloid sheets)	(2) $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	Two disconnected pieces

5 Classification Criteria

Given a quadric $F(x, y, z) = 0$, the standard classification proceeds as follows.

1. **Degree of equation.** If F is linear ($\deg F = 1$), the surface is a *plane*.
2. **Signs of squared-term coefficients.** After reducing to standard form, count the positive (p) and negative (q) squared terms:
 - $p = 3, q = 0$: ellipsoid (sphere if all coefficients equal).
 - $p = 2, q = 1$: hyperboloid of one sheet or cone ($= 0$ RHS).
 - $p = 1, q = 2$: hyperboloid of two sheets.
 - $p = 2, q = 0$ with one linear term: elliptic/circular paraboloid.
 - $p = 1, q = 1$ with one linear term: hyperbolic paraboloid.
3. **Missing variable.** If F is independent of one coordinate (say z), the surface is a *cylinder* whose cross-section is determined by the two-variable curve $F(x, y) = 0$.

4. **Right-hand side.** For surfaces of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = k$: $k > 0$ gives one sheet, $k = 0$ gives a cone, $k < 0$ gives two sheets.

6 Conclusion

Surfaces in $\mathbb{R}^3$ range from the simplest flat *plane* (degree 1) to the rich family of *quadric surfaces* (degree 2). The key classification criteria are: the degree of the defining polynomial, the signs of the squared-variable coefficients, whether a variable is absent (cylindrical surface), and the value of the constant term. Understanding these distinctions is fundamental in multivariable calculus, differential geometry, and linear algebra.